

Getting started with Google Colab

A step-by-step guide to running the Class 1 notebooks — written for people who have never coded.

Who this is for — everyone, especially if you have never written a line of code · **Time needed** — about 20 minutes, once · **You need** — a laptop, a web browser, and internet. Nothing to install.

Two-thirds of this class arrived without a technical background — you are the majority, not the exception. Google Colab lets you run real code inside your web browser, with nothing to download and nothing to set up on your computer. This guide walks you through it slowly, one click at a time. If a step doesn't look exactly like the picture in your head, that is normal — read the checkpoint boxes to confirm you are on track, and use the troubleshooting table at the end.

WHAT IS GOOGLE COLAB?

A free Google service that runs code on Google's computers, not yours. You open it in a browser tab — like opening Gmail — and everything happens there. The files we run are called **notebooks** (they end in `.ipynb`). A notebook is just text and code broken into **cells** that you run one at a time. That's the whole idea.

Part A · Create a Google account (skip if you already have one)

Colab is a Google service, so you sign in with a Google account. If you already use Gmail, YouTube, or an Android phone, you already have one — **skip to Part B**. If not, making one takes five minutes and is free.

- 1 Open your web browser (**Google Chrome** or **Microsoft Edge** work best) and go to `accounts.google.com/signup`
- 2 Enter your **name**, then choose a **username** — this becomes your email address, for example `yourname@gmail.com`. Pick a **password** you will remember.
- 3 Google may ask for a **phone number** to confirm you are a real person. A mainland China (+86), Singapore (+65), or any country's mobile number all work. You will receive a text message with a code — type it in.
- 4 Accept the terms. That's it — you now have a Google account and are signed in.

IMPORTANT — IF YOU ARE FROM MAINLAND CHINA

Google works normally on the **NTU campus wifi and anywhere in Singapore** — you do not need anything special here. You only run into trouble if you try to open Colab from *inside mainland China* (for example during a trip home), where Google is blocked and you would need a VPN. For all of Class 1 and the coursework, you will be in Singapore, so this will not affect you. Just create the account once, now, while you are on campus.

✓ CHECKPOINT

You can open `mail.google.com` and see an inbox, or you see your name / photo in the top-right circle of any Google page. That means you are signed in.

Part B · Open Colab and run your first cell

Before touching the class notebooks, let's make sure Colab works for you with a tiny test.

- 1 In your browser, go to `colab.research.google.com`
- 2 If a pop-up window appears listing notebooks, click **New notebook** at the bottom. (If you are asked to sign in, use the Google account from Part A.)
- 3 You will see an empty grey box — that is a **cell**. Click inside it and type exactly: `print("Hello Colab")`
- 4 Press **Shift** + **Enter** together to run the cell. The first run takes 10–20 seconds while Colab wakes up a computer for you — this wait only happens once per session.

✓ CHECKPOINT

The words `Hello Colab` appear just below the cell. If you see that, Colab is working and you are ready for the real notebooks. You can close this test tab.

GOOD TO KNOW

Everything you do in Colab is temporary until you save it (Part F). If you leave a notebook idle for a long time, Colab disconnects the computer to save resources — your code stays, you just press run again. Nothing breaks.

Part C · Get the Class 1 notebooks and open them

There are **two notebooks** for Class 1. You download them from NTULearn to your laptop, then upload each one into Colab.

Notebook file	What it shows
<code>PE6201_C2_classifier_vs_api.ipynb</code>	The same task done two ways — a trained classifier vs. a foundation-model API. "One needs your data, the other needs your money."
<code>PE6201_C3_tokenizer_demo.ipynb</code>	Why a model can't count the letters in "strawberry" — tokens, and why some languages cost more.

- 1 Log in to **NTULearn** and open the PE6201 Class 1 materials. Download the two `.ipynb` files above to your laptop (they will land in your **Downloads** folder). Do not try to open them by double-clicking — they only open properly inside Colab.
- 2 Go back to `colab.research.google.com`. In the pop-up, click the **Upload** tab at the top, then **Choose file** (or drag the file in).
- 3 Pick `PE6201_C2_classifier_vs_api.ipynb` from your Downloads. It opens in a new Colab tab, ready to run.
- 4 To open the second notebook later, use the menu **File ► Upload notebook** and choose `PE6201_C3_tokenizer_demo.ipynb`.

✓ CHECKPOINT

You see the notebook's title at the top and a series of cells down the page, some with text and some with code. Nothing has run yet — that's next.

Part D · Run a notebook

You have two ways to run a notebook. Both are fine — pick whichever feels comfortable.

The easy way — run everything at once

- 1 Click the menu **Runtime ► Run all**.
- 2 If a box warns "*This notebook was not authored by Google*", click **Run anyway**. This is normal — it just means the notebook came from us, not from Google.
- 3 Watch the cells run top to bottom. A small spinning circle means a cell is working; a number in `[ ]` brackets means it finished. Results and colourful tables appear underneath each cell.

The learn-as-you-go way — one cell at a time

Click the first cell, press `Shift` + `Enter` to run it and move to the next. Repeat. This lets you read what each cell does before running it — recommended if you want to actually follow the lesson.

TWO THINGS YOU WILL SEE — BOTH ARE NORMAL

1. A line starting with `!pip install tiktoken` (in the tokenizer notebook). This downloads a small tool the notebook needs. It takes a few seconds and prints some grey text — let it finish.
2. A pause of up to a minute on the classifier notebook when it trains on 10,000 examples. The computer is doing real work. Wait for the `[ ]` to show a number.

✓ CHECKPOINT

In the classifier notebook you should end up seeing a comparison table ("Trained classifier" vs "Foundation model"). In the tokenizer notebook you should see the word "strawberry" broken into coloured blocks. If you see those, the notebook ran correctly.

Part E · The API key and the RUN_LIVE switch

An **API [Application Programming Interface] key** is like a password that lets a notebook call a live AI model over the internet — and each call costs a small amount of real money. Every notebook has one switch that decides whether the key is used, called `RUN_LIVE`. You only ever change two things: this switch, and where the key goes.

`RUN_LIVE = False`

Does **not** use the API key. The notebook runs on safe stand-in results, for free. **This is Class 1** — leave it here.

`RUN_LIVE = True`

Uses your API key to call a real AI model. **This is Assignment 1**, once you have your own key.

✓ FOR CLASS 1

Do nothing. Both Class 1 notebooks already have `RUN_LIVE = False` and run fully without a key. The rest of this page is for **Assignment 1**, when you will have your own key.

How you will get your key

For **Assignment 1**, you will be given your **own personal API key** with credit already loaded — released on NTU Learn together with the starter notebooks. It is a long line of text that starts with `sk-or-v1-` followed by many letters and numbers. **Treat it like a password: never share it, and never post it in a chat or email.**

Turn the switch on and save your key into the notebook

Near the top of the assignment notebook there is a small **settings cell**. You edit just two lines in it, shown below.

```
# --- Settings: change these two lines ---
```

```
RUN_LIVE = True 1
```

```
API_KEY = "sk-or-v1-...paste yours..." 2
```

```
MODEL = "openai/gpt-4o-mini"
```

The settings cell. The round button on the left runs the cell. You only touch the two highlighted lines.

- 1 Change `RUN_LIVE = False` to `RUN_LIVE = True`. (True = use the AI model; False = do not.)
- 2 Click **between the two quotation marks** on the `API_KEY` line, delete what is there, and paste your own key. **Keep the quotation marks** around it.

Then run that cell (click it, press **Shift** + **Enter**) and continue with the notebook. If the key is saved correctly, the live cells now return real answers instead of the stand-in note.

BEFORE YOU SUBMIT OR SHARE — REMOVE YOUR KEY

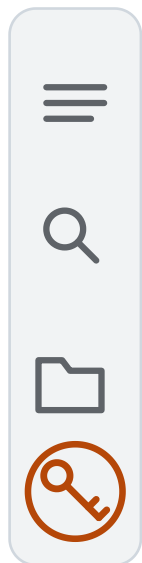
You hand in your notebook file, and your key is personal to you. Before you download and submit, change the `API_KEY` line back to empty — `API_KEY = ""` — and set `RUN_LIVE = False`. This stops your key from being copied along with your work.

Optional: the safer way — Colab "Secrets"

Colab has a built-in vault for keys called **Secrets**, opened from the **key icon** on the far-left strip of the screen. Storing your key there keeps it out of the notebook file entirely, so you never have to remove it before submitting. Only use this if your starter notebook mentions "Secrets" or `userdata` — otherwise stick with the two steps above.

IMPORTANT — THE NAME MUST MATCH

When you add a secret, you give it a **name**. That name must be **spelled exactly the same** as the one the notebook looks for — for example, if the notebook reads `userdata.get("OPENROUTER_API_KEY")`, your secret must be named `OPENROUTER_API_KEY`, letter for letter (capitals included). If the names differ, the notebook will not find your key.



Files

where your uploaded notebook lives

Secrets — the key icon

optional safe storage for your API key

The strip of icons down the left edge of Colab. The key icon opens Secrets.

Part F · Save your work and hand it in

- 1 To keep your work, click **File ► Save a copy in Drive**. This saves the notebook to your own Google Drive so you can reopen it anytime from colab.research.google.com under the **Recent** tab.
- 2 Colab also **auto-saves** to your Drive as you work — but clicking save yourself is the safe habit.
- 3 When an assignment asks you to submit a notebook, use **File ► Download ► Download .ipynb** to get the file onto your laptop, then upload that file to NTULearn.

Troubleshooting — if something looks wrong

What you see	What to do
" Cannot connect " / the page won't load Google at all	Check you are on NTU campus wifi or normal Singapore internet. If you are travelling in mainland China, Google is blocked there and you would need a VPN.
" This notebook was not authored by Google "	Normal. Click Run anyway . The notebook came from the course, which Google can't verify — it is safe.
A cell shows a red error message	Run the cells from the very top, in order — the most common cause is running a later cell before an earlier one. Use Runtime ► Run all to be sure.
" Your session crashed " or it disconnected	Click Runtime ► Restart session , then Runtime ► Run all again. Your code is safe; only the running computer restarted.

<code>NameError</code> — something "is not defined"	Same fix: a cell that defines it hasn't been run yet. Run everything from the top.
The <code>!pip install</code> line seems stuck	Give it up to a minute. It prints grey text while downloading. It is done when the <code>[]</code> shows a number.
It's asking me for an API key	For Class 1 you don't need one — set <code>RUN_LIVE = False</code> and re-run the cell. For Assignment 1 , set <code>RUN_LIVE = True</code> and paste your key (Part E).
I lost my notebook	Open <code>colab.research.google.com</code> and look under the Recent or Google Drive tab — if you saved it (Part F), it's there.

One-page cheat sheet

To do this	Do this
Open Colab	Go to <code>colab.research.google.com</code> and sign in with your Google account
Open a class notebook	File ► Upload notebook → choose the <code>.ipynb</code> you downloaded from NTULearn
Run one cell	Click the cell, press <code>Shift</code> + <code>Enter</code>
Run the whole notebook	Runtime ► Run all
Start fresh if stuck	Runtime ► Restart session , then Run all
Use the AI model (Assignment 1)	Set <code>RUN_LIVE = True</code> , paste your key into <code>API_KEY = ""</code> (set back to <code>False</code> / <code>""</code> before submitting)
Save your work	File ► Save a copy in Drive
Get a file to submit	File ► Download ► Download .ipynb

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